

1	Course Name:
	Digital Principles
2	Course Code:
	IBB30104
3	Year and Semester Offered:
	Year 2 Semester 1
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NIL
6	Course Synopsis:
	This course covers the core fundamentals of digital technology. The concepts range from Digital and analog quantities to Numbering Systems to Logic Gates, Boolean Algebra, and logic simplification. Combinational design techniques as well as sequential design techniques are also presented with the use of Karnaugh mapping. The course will also go through the functions of combinational logic, latches and flip flops, and shift registers and counters.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Explain fundamental concepts of digital principles including basic and universal gates, number systems and binary coded systems. 2. Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital circuits including Boolean algebra and Karnaugh map methods. 3. Analyze small-scale combinational and sequential digital circuits. 4. Construct digital circuits using laboratory equipments or simulation tools.
8	Content Outline:
	Topic
	1. Introductory Concepts
	2. Number Systems, Operations and codes
	3. Logic Gates
	4. Boolean Algebra and Logic Simplification
	5. Combinational Logic Analysis
	6. Functions of Combinational Logic
	7. Latches, Flip-Flops
	8. Counters & Shift Register
9	References:
	Roger L. Tokheim and Patrick E. Hoppe, Digital Electronics: Principles and Applications, 9th Edition, 2021, ISBN-10: 1260597865